

FLEET RULE OF THE ROAD TEST OCTOBER 2014

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(E)Vol I and the Mariners Handbook (NP100).

BR454 "A Guide to the Collision Avoidance Rules" 6th Edition by Cockcroft and Lameijer was categorised as obsolescent by NCHQ in August 2011. Care should be taken when utilising the 6th Edition as the information contained within may not reflect the most recent changes to the regulations.

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the ROR/training logbook for inspection by FLEET/FOST navigation staff.

In this test, any change in the status of the ship or of the environment will be indicated in the script.

Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

Answers are provided for uniformity of marking. This test is completely scenario based.

PART 1

You are the SSD Officer of the Watch of a T45 destroyer (length 152 meters). You are departing Portsmouth Harbour and have just passed Spit Sand Fort abeam to starboard on the 181 Leg towards OSB. It is the forenoon watch and rain showers have reduced visibility to 4NM. The shipping is moderate.

1. You sight a small power-driven vessel at R20, range 1NM. You take an initial bearing and notice that you are 10 degrees on the starboard bow of the vessel. The DMEO asks if the Rules apply to this vessel. What is your answer and to whom do they apply? **(R. 1a)** (2)
2. You report the vessel to the Captain who observes that the contact has made a bold alteration of course to starboard. The bearing of the vessel is now drawing safely left and passes 5 cables to port. As you are navigating in a narrow channel, which vessels should not impede your safe passage? **(R. 9b)** (2)
3. How should you proceed along the course of a narrow channel or fairway? **(R. 9a)** (1)

4. As you pass OSB to starboard you notice that a hovercraft has departed Ryde Pier, now at G50, range 3NM. At night, when making way, what lights would you expect the hovercraft (>50m length) to display? **(R. 23b)** (3)

5. You make an alteration of course to port and continue towards Nab Channel. Operating close to St.Helen's Road Anchorage, you notice a small group of vessels engaged in fishing. How are these vessels defined by the Rules? **(R. 3d)** (2)

6. As the vessels are operating by day, what shapes should they exhibit? **(R. 26c)** (1)

7. You continue your passage along the channel and sight the Motor Tanker Hanne Knutsen inbound to Fawley. You are on her Ship's head and her bearing is steady. In this situation:

a. What action should you take in accordance with the Rules? **(R. 14a)** (1)

b. When shall such a situation be deemed to exist? **(R. 14b)** (2)

c. You inform the Captain that you intend to manoeuvre as required by Rule 14. He asks you to make the appropriate signal using the Ship's whistle. What sound signal do you make? **(R. 34a)** (1)

8. As you pass the Hanne Knutsen safely, you notice that a black cylinder is displayed at the masthead. What does this day shape indicate? **(R. 28)** (1)

a. How is such a vessel defined by the Rules? **(R. 3h)** (2)

b. How should the Hanne Knutsen navigate in the channel? **(R. 18d)** (1)

- c. If the circumstances of the case admit, which vessels should avoid impeding the safe passage of a vessel constrained by her draught? **(R. 18d)** (3)
- d. What lights would you expect this vessel to exhibit at night and what is the minimum range you would expect to see them? **(R. 22/28)** (3)
9. You continue past Nab Tower and the traffic density begins to reduce. On falling out from SSD you are concerned by the deteriorating weather in the vicinity. A heavy rainstorm passes through and visibility reduces significantly. You call the CO and NO to inform them. In doing so you have determined that in the prevailing situation, safe speed is 10 knots.
- a. In determining safe speed what factors shall be among those taken into account by all vessels? **(R. 6a)** (3)
- b. What sound signal should you make? **(R. 35a)** (1)
- c. What do the rules state about safe speed in restricted visibility? **(R. 19b)** (1)
10. You receive a call from the SPS who reports a contact on radar at G30 range 4 miles. The CPA is reported as 3 cables on your starboard beam.
- a. What do the rules state about this situation and the actions you should take? **(R. 19d)** (2)
- b. You make a shipping report to the CO and take avoiding action in good time. Continuing south-west towards St. Catherine's Point, you hear a fog signal to port consisting of one prolonged blast followed by two short blasts. What type of vessel could this be? **(R. 35c)** (3)

c. You check the radar and hold no steady contacts on the port side other than a faint echo at R20, range 5 cables. You cannot track the echo effectively or generate an accurate CPA using the radar. What action should you take in accordance with the rules? **(R. 19e) (2)**

d. The contact closes to 1 mile and you sight the following lights:

R
G

R

e. What type of vessel is this? **(R. 25a/c) (2)**

11. The vessel passes clear down your port side and you notice that visibility is rapidly improving as the rainstorm passes through to the north. You assess visibility as 10nm and inform the CO who is happy for you now to increase speed to meet the required SOA. As you come up in speed you sight a small cargo vessel at G30, range 3nm. You assess that you are 40 degrees on her port bow and her bearing is steady.

a. In accordance with the Rules what action should you take? **(R. 15) (1)**

b. What should the cargo vessel do? **(R. 17) (1)**

c. The vessel is now at 5 cables and you have not taken any action to avoid collision. You decide, given the closing range, that urgent action is required to avoid collision and inform the CO. What could you decide to do? **(R. 8e) (1)**

d. How should action be taken to avoid collision in accordance with the rules? **(R. 8a) (1)**

12. The cargo vessel takes avoiding action in accordance with Rule 17b and makes a bold alteration of course to starboard. A collision narrowly avoided, the vessel passes 2 cables clear down your port side. The CO comes to the bridge to debrief you on the recent incident. He stresses the importance of the Rules and the requirement to act early to generate 'thinking time' in a developing situation. You both reflect on the shipping situation and agree that a simple alteration of course would have resolved the problem early. If taking action to avoid collision by alteration of course alone, what should you consider? **(R. 8c)** (2)

a. How apparent should the alteration of course be and what should you avoid doing? **(R. 8b)** (2)

13. The CO calls the NO and instructs him to plan a series of OOW shareholders to discuss RoR and likely scenarios in more detail. The Captain would like you to focus on Rules 8 and 18. Your 2OOW is interested in responsibility between vessels and asks what is required of the Ship. What is your answer? **(R. 18a)** (2)

You are relieved by the NO who is keen to take a watch as the afternoon programme IAW the WPP is fairly uncomplicated. You sigh in relief as you leave the bridge and return to your cabin before lunch.

Total 49 marks

ANSWERS

PART 1

You are the SSD Officer of the Watch of a T45 destroyer (length 152 meters). You are departing Portsmouth Harbour and have just passed Spit Sand Fort abeam to starboard on the 181 Leg towards OSB. It is the forenoon watch and rain showers have reduced visibility to 4NM. The shipping is moderate.

1. You sight a small power-driven vessel at R20, range 1NM. You take an initial bearing and notice that you are 10 degrees on the starboard bow of the vessel. The DMEO asks if the Rules apply to this vessel. What is your answer and to whom do they apply? **(R. 1a) (2)**

Yes. (R. 1a) These rules shall apply to all vessels upon the high seas and in all waters connected therewith navigable by seagoing vessels.

2. You report the vessel to the Captain who observes that the contact has made a bold alteration of course to starboard. The bearing of the vessel is now drawing safely left and passes 5 cables to port. As you are navigating in a narrow channel, which vessels should not impede your safe passage? **(R. 9b) (2)**

(R.9b) (b) A vessel of less than 20 meters in length or a sailing vessel shall not impede the passage of a vessel which can safely navigate only within a narrow channel or fairway.

3. How should you proceed along the course of a narrow channel or fairway? **(R. 9a) (1)**

(R. 9a) A vessel proceeding along the course of a narrow channel or fairway shall keep as near to the outer limit of the channel or fairway which lies on her starboard side as is safe and practicable.

4. As you pass OSB to starboard you notice that a hovercraft has departed Ryde Pier, now at G50, range 3NM. At night, when making way, what lights would you expect the hovercraft (>50m length) to display? **(R. 23b) (3)**

(R. 23b)

(a) A power-driven vessel underway shall exhibit:

(i) a masthead light forward;

(ii) a second masthead light abaft of and higher than the forward one;

except that a vessel of less than 50 meters in length shall not be obliged to exhibit such a light but may do so;

(iii) sidelights: and

(iv) a sternlight.

(b) An air-cushion vessel when operating in non-displacement mode shall, in addition to the lights prescribed in paragraph (a) of this Rule, exhibit an all-round flashing yellow light.

5. You make an alteration of course to port and continue towards Nab Channel. Operating close to St. Helen's Road Anchorage, you notice a small group of vessels engaged in fishing. How are these vessels defined by the Rules? **(R. 3d) (2)**

(R. 3d) The term "vessel engaged in fishing" means any vessel fishing with nets, lines, trawls, or other fishing apparatus which restrict manoeuvrability, but does not include a vessel fishing with trolling lines or other fishing apparatus which do not restrict manoeuvrability .

6. As the vessels are operating by day, what shapes should they exhibit? **(R. 26c) (1)**

(R. 26c) A shape consisting of two cones with their apexes together in a vertical line one above the other; a vessel of less than 20 meters in length may instead of this shape exhibit a basket.

7. You continue your passage along the channel and sight the Motor Tanker Hanne Knutsen inbound to Fawley. You are on his Ship's head and his bearing is steady. In this situation:

c. What action should you take in accordance with the Rules? **(R. 14a) (1)**

(R.14a) Head-on Situation

(a) When two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision each shall alter her course to starboard so that each shall pass on the port side of the other.

d. When shall such a situation be deemed to exist? **(R. 14b) (2)**

(R. 14b) Such a situation shall be deemed to exist when a vessel sees the other ahead or nearly ahead and by night she could see the masthead lights in line or nearly in line and/or both sidelights and by day she observes the corresponding aspect of the other vessel.

c. You inform the Captain that you intend to manoeuvre as required by Rule 14. He asks you to make the appropriate signal using the Ship's whistle. What sound signal do you make? **(R. 34a) (1)**

(R. 34a) One short blast to mean "I am altering my course to starboard".

8. As you pass the Hanne Knutsen safely to port, you notice that a black cylinder is displayed at the masthead. What does this day shape indicate? **(R. 28) (1)**

(R. 28) Vessels Constrained by their Draft

A vessel constrained by her draft may, in addition to the lights prescribed for power-driven vessels in Rule 23, exhibit where they can best be seen three all-round red lights in a vertical line, or a cylinder.

c. How is such a vessel defined by the Rules? **(R. 3h) (2)**

(R. 3h) The term "vessel constrained by her draft" means a power-driven vessel which because of her draft in relation to the available depth and width of navigable water is severely restricted in her ability to deviate from the course she is following.

d. How should the Hanne Knutsen navigate in the channel? **(R. 18d) (1)**

(R. 18d) A vessel constrained by her draft shall navigate with particular caution having full regard to her special condition.

c. If the circumstances of the case admit, which vessels should avoid impeding the safe passage of a vessel constrained by her draught? **(R. 18d) (3)**

(R. 18d) Any vessel other than a vessel not under command or a vessel restricted in her ability to manoeuvre shall, if the circumstances of the case admit, avoid impeding the safe passage of a vessel constrained by her draft.

d. What lights would you expect this vessel to exhibit at night and what is the minimum range you would expect to see them? **(R. 22/28) (3)**

(R. 22/23/28) A vessel constrained by her draft may, in addition to the lights prescribed for power-driven vessels in Rule 23, exhibit where they can best be seen three all-round red lights in a vertical line, or a cylinder.

A power-driven vessel underway shall exhibit:

- (i) a masthead light forward;**
- (ii) a second masthead light abaft of and higher than the forward one; except that a vessel of less than 50 meters in length shall not be obliged to exhibit such a light but may do so;**
- (iii) sidelights: and**
- (iv) a sternlight.**

(In vessels of 50 meters or more in length:

Masthead light, 6 miles;

Sidelight, 3 miles;

Towing light, 3 miles;

White red, green or yellow all-around light, 3 miles.

9. You continue past Nab Tower and the traffic density begins to reduce. On falling out from SSD you are concerned by the deteriorating weather in the vicinity. A heavy rainstorm passes through and visibility reduces significantly. You call the CO and NO to inform them. In doing so you have determined that in the prevailing situation, safe speed is 10 knots.

a. In determining safe speed what factors shall be among those taken into account by all vessels? **(R. 6a) (3)**

(R.6a) By all vessels:

- (i) The state of visibility;**
- (ii) The traffic density including concentrations of fishing vessels or any other vessels;**
- (iii) The manageability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions;**
- (iv) At night the presence of background light such as from shore lights or from back scatter from her own lights;**
- (v) The state of wind, sea and current, and the proximity of navigational hazards;**
- (vi) The draft in relation to the available depth of water.**

b. What sound signal should you make? **(R. 35a) (1)**

(R. 35a) A power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes one prolonged blast.

c. What do the rules state about safe speed in restricted visibility?
(R. 19b) (1)

(R. 19b) Every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions of restricted visibility. A power-driven vessel shall have her engines ready for immediate manoeuvre.

10. You receive a call from the SPS who reports a contact on radar at G30 range 4 miles. The CPA is reported as 3 cables on your starboard beam.

- a. What do the rules state about this situation and the actions you should take? **(R. 19d) (2)**

(R. 19d) A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists. If so, she shall take avoiding action in ample time, provided that when such action consists of an alteration in course, so far as possible the following shall be avoided:

(i) An alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;

(ii) An alteration of course toward a vessel abeam or abaft the beam.

- b. You make a shipping report to the CO and take avoiding action in good time. Continuing south-west towards St. Catherine's Point, you hear a fog signal to port consisting of one prolonged blast followed by two short blasts. What type of vessel could this be? **(R. 35c) (3)**

(R. 35c) A vessel not under command, a vessel restricted in her ability to manoeuvre, a vessel constrained by her draft, a sailing vessel, a vessel engaged in fishing and a vessel engaged in towing or pushing another vessel shall, instead of the signals prescribed in paragraph (a) or (b) of this Rule, sound at intervals of not more than 2 minutes sound three blasts in succession, namely one prolonged followed by two short blasts.

- c. You check the radar and hold no steady contacts on the port side other than a faint echo at R20, range 5 cables. You cannot track the echo effectively or generate an accurate CPA using the radar. What action should you take in accordance with the rules? **(R. 19e) (2)**

(R. 19e) Except where it has been determined that a risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close-quarters situation with another vessel forward of her beam, shall reduce her speed to be the minimum at which she can be kept on her course. She shall if necessary take all her way off and in any event navigate with extreme caution until danger of collision is over.

- d. The contact closes to 1 mile and you sight the following lights:

R
G

R

- e. What type of vessel is this? **(R. 25a/c) (2)**

(R. 25a/c) A sailing vessel underway shall exhibit:

- (i) sidelights;**
(ii) a sternlight.

A sailing vessel underway may, in addition to the lights prescribed in paragraph (a) of this Rule, exhibit at or near the top of the mast, where they can best be seen, two all-

round lights in a vertical line, the upper being red and the lower green, but these lights shall not be exhibited in conjunction with the combined lantern permitted by paragraph (b) of this Rule.

11. The vessel passes clear down your port side and you notice that visibility is rapidly improving as the rainstorm passes through to the north. You assess visibility as 10nm and inform the CO who is happy for you now to increase speed to meet the required SOA. As you come up in speed you sight a small cargo vessel at G30, range 3nm. You assess that you are 40 degrees on her port bow and her bearing is steady.

- a. In accordance with the Rules what action should you take? **(R. 15) (1)**

(R. 15) Crossing Situation

When two power-driven vessels are crossing so as to involve risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel.

- b. What should the cargo vessel do? **(R. 17) (1)**

(R. 17) Where one of two vessels is to keep out of the way, the other shall keep her course and speed.

- c. The vessel is now at 5 cables and you have not taken any action to avoid collision. You decide, given the closing range, that urgent action is required to avoid collision and inform the CO. What could you decide to do? **(R. 8e) (1)**

(R. 8e) If necessary to avoid collision or allow more time to assess the situation, a vessel may slacken her speed or take all way off by stopping or reversing her means of propulsion.

- d. How should action be taken to avoid collision in accordance with the rules? **(R. 8a) (1)**

(R. 8a) Any action to avoid collision shall be taken in accordance with the Rules of this Part and shall, if the circumstances of the case admit, be positive, made in ample time and with due regard to the observance of good seamanship.

12. The cargo vessel takes avoiding action in accordance with Rule 17b and makes a bold alteration of course to starboard. A collision narrowly avoided, the vessel passes 2 cables clear down your port side. The CO comes to the bridge to debrief you on the recent incident. He stresses the importance of the Rules and the requirement to act early to generate 'thinking time' in a developing situation. You both reflect on the shipping situation and agree that a simple alteration of course would have resolved the problem early. If taking action to avoid collision by alteration of course alone, what should you consider? **(R. 8c) (2)**

(R. 8c) If there is sufficient sea room, alteration of course alone may be the most effective action to avoid a close-quarters situation provided that it is made in good time, is substantial and does not result in another close-quarters situation.

- a. How apparent should the alteration of course be and what should you avoid doing? **(R. 8b) (2)**

(R. 8b) Any alteration of course and/or speed to avoid collision shall, if the circumstances of the case admit be large enough to be readily apparent to another vessel observing

visually or by radar; a succession of small alterations of course and/or speed shall be avoided.

13. The CO calls the NO and instructs him to plan a series of OOW shareholders to discuss RoR and likely scenarios in more detail. The Captain would like you to focus on Rules 8 and 18. Your 2OOW is interested in responsibility between vessels and asks what is required of the Ship. What is your answer? **(R. 18a)** (2)

(R. 18a) Except where rule 9, 10, and 13 otherwise require:

(A power driven vessel underway shall keep out of the way of:

- (i) A vessel not under command;**
- (ii) A vessel restricted in her ability to maneuver;**
- (iii) A vessel engaged in fishing;**
- (iv) A sailing vessel.**

You are relieved by the NO who is keen to take a watch as the afternoon programme IAW the WPP is fairly uncomplicated. You sigh in relief as you leave the bridge and return to your cabin before lunch.

Total 49 marks